

will purchase those shares because they are outside the specified price range. The result: lower trading volume. The new equilibrium point will be the intersection of the original demand curve and the new supply curve S' , in agreement with a new equilibrium price of p_4 and a new equilibrium quantity of q_{new} . Here we expect a post-trade price increase ($p_4 > q^*$) and a post-trade volume decrease ($q_{\text{new}} < q^*$). A breakdown of the market impact cost in this scenario is total impact = $p_2 - p^*$, with temporary impact = $p_2 - p_4$ and permanent impact = $p_4 - p^*$.

In a second scenario the original number of buyers may continue to purchase the same number of shares even at higher prices. For example, index managers hold certain stocks and quantities in their portfolio regardless of their market prices because they need to mimic the underlying index. Therefore, after the increment shares Δq are transacted the number of buyers returns to pre-trade levels. Since they are willing to transact at higher prices the demand curve returns to a higher level D'' . The new equilibrium point: the point of intersection between S' and D'' . Demand is identical to the pre-trade level q^* while the price will be p_3 (higher than the original equilibrium level of p^* and also higher than p_4 (from the first scenario) where we assumed fewer buyers post-trade due to the higher prices). A breakdown of the market impact cost in this scenario is total impact = $p_2 - p^*$, with temporary impact = $p_2 - p_3$ and permanent impact = $p_3 - p^*$. In both scenarios, the total impact of the trade is identical except for new post-trade equilibrium points. This results in computations for permanent and temporary impact along with different expectations for forward looking market volumes.

New equilibrium demand level and price uncertainty are major reasons behind the difficulty in distinguishing between temporary and permanent market impact cost. Regrettably, this is rarely addressed in the financial literature.

The question remains: Does excess market volume lead to more or less volume going forward? We often find that excess market volume corresponds with excess volume in the short-term. However, the higher volume is generally attributed to news—for example, earnings, major macro-economic events or new announcements, corporate actions, and so on. Higher volume can also tie to investors implementing substantial orders executed over multiple days. We have uncovered evidence of volume returning to its original state as well as volume levels returning to lower levels. No statistical evidence exists suggesting that levels would remain at a higher state. In the rare cases where volume stood higher than pre-trade levels, the reasoning was: (1) the stock joined an index, (2) merger